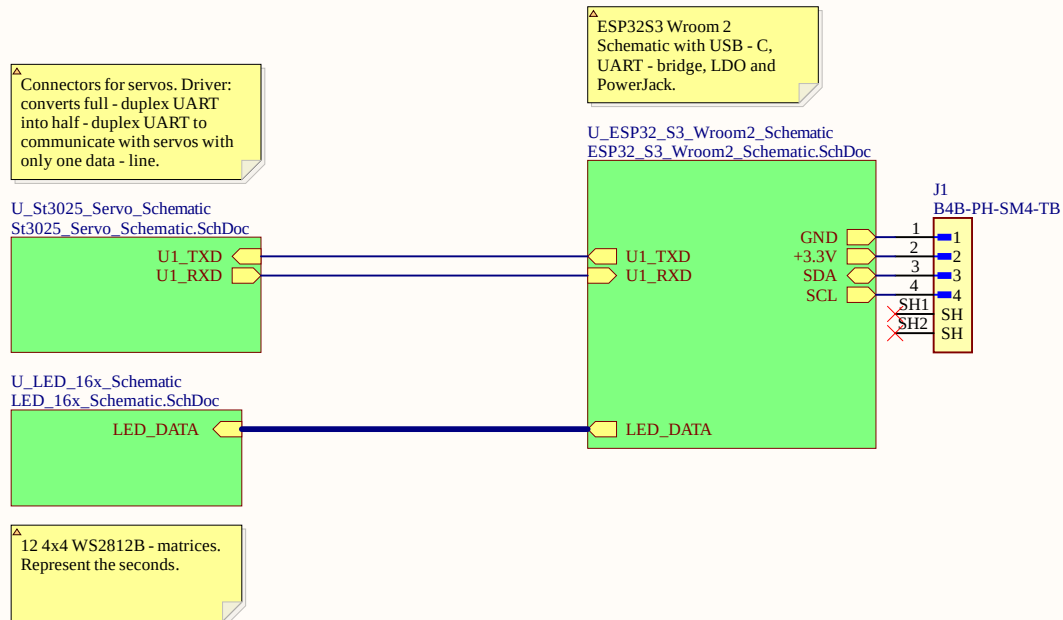
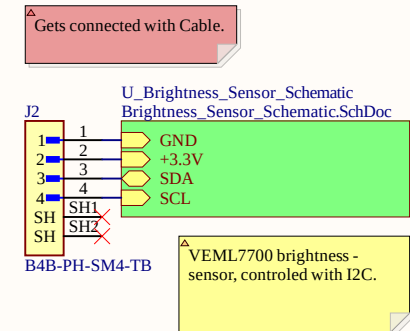


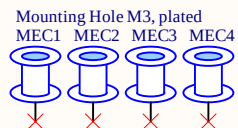
Main - BauhallenUhr - Board



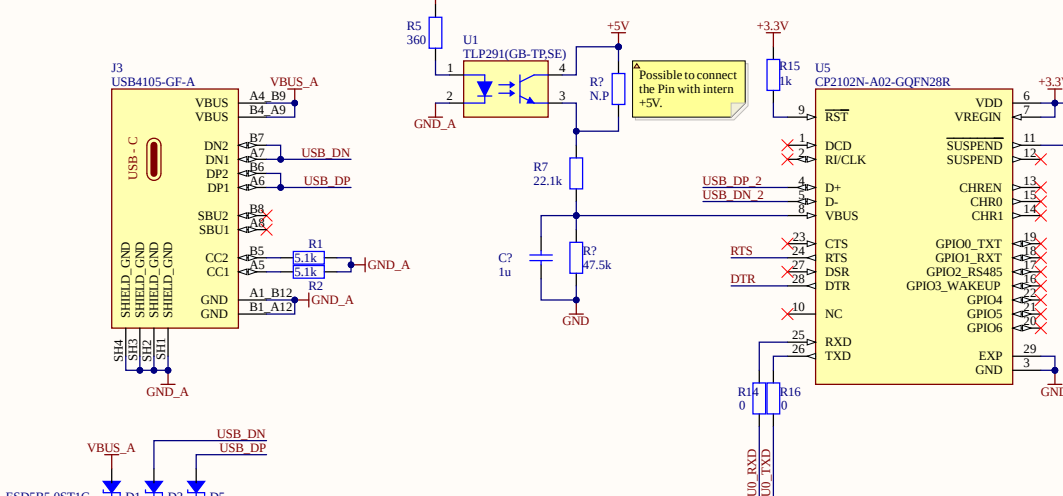
Brightness - Sensor - Board



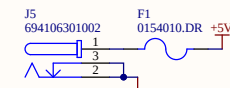
Mounting Holes



USB to UART



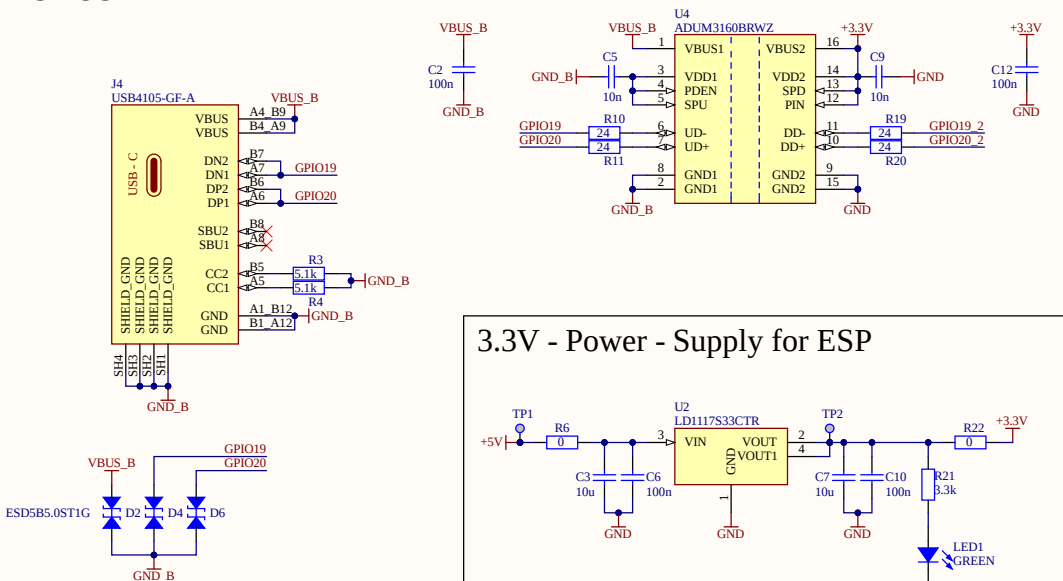
Power - Supply



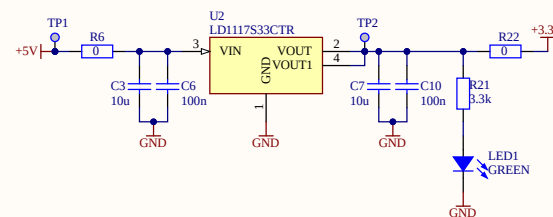
I2C Connector Entries



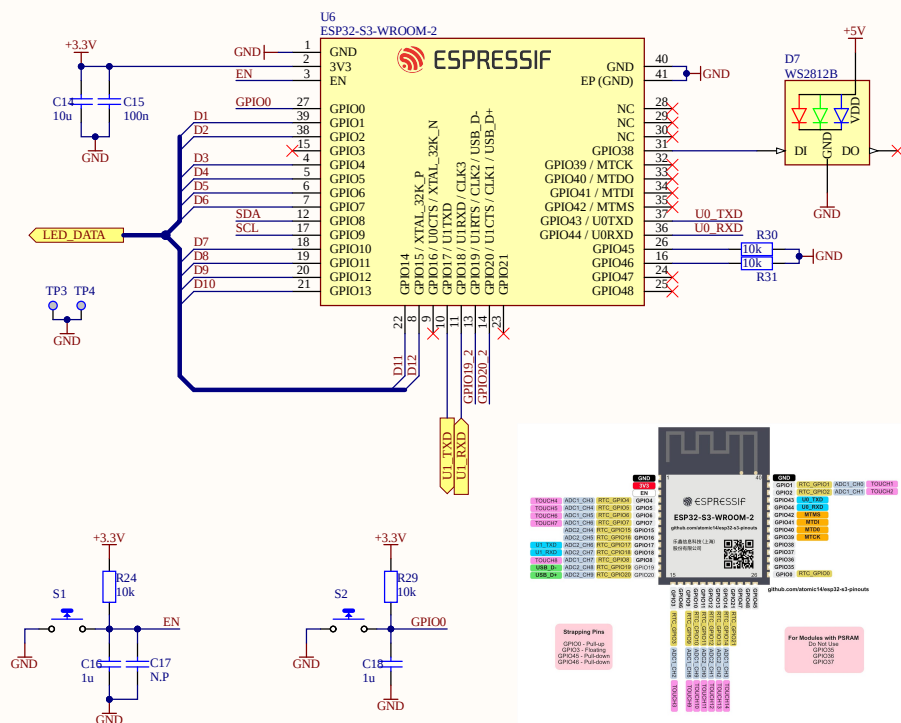
ESP USB



3.3V - Power - Supply for ESP



ESP32S3 Wroom 2 Module



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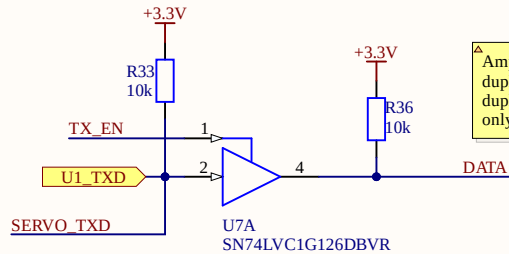
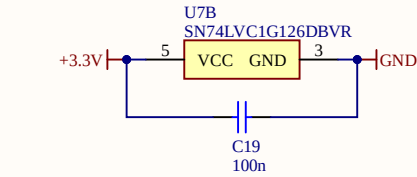


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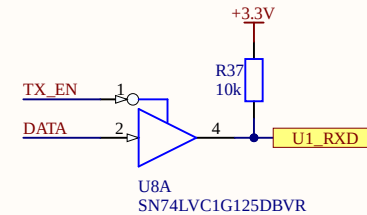
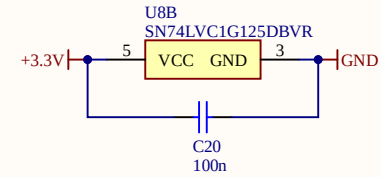
Date: 30.07.2026 Time: 10:04:49
Revision: V0.1
Drawn by: Manuel Hueppi
Institut: IBK
Department: DBAUG
Project: BauhallenUhr.PrjPcb
Page: 2 from: 6

File: C:\Users\hueppi\Documents\Github\BauhallenUhr\Altium_BauhallenUhr_V0.1\ESP32_S3_Wroom2_Schematic.SchDoc

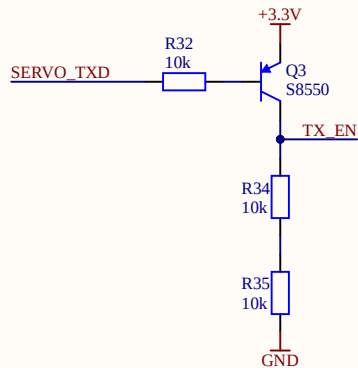
Single Bus Buffer Gate



Inverted Bus Buffer Gate



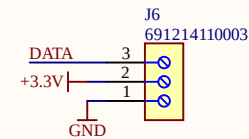
Half - Duplex Direction Detection



If the ESP32 writes something to the servos, the TX_EN - signal will enable the data flow between the SERVO_TXD - line and the DATA - line. If the ESP32 writes nothing, the data flow in the opposite direction is enabled, which means the ESP32 can read e. g the position of the servos.

Servo Connector

Only one connector, because these servos are controlled via UART, they can be chained together (serial).



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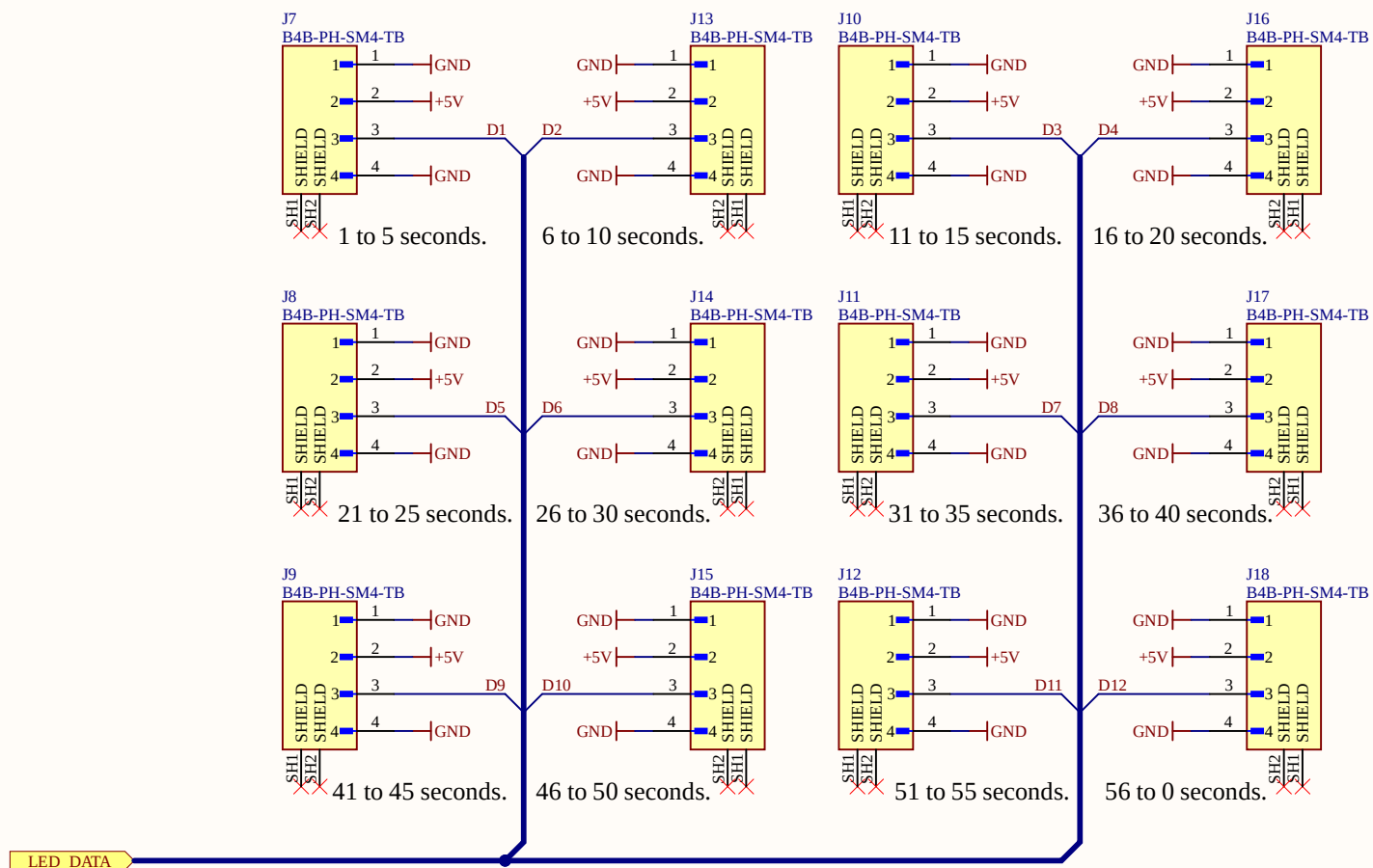
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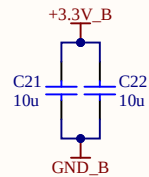
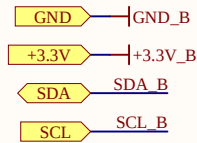
Drawn by: Manuel Hueppi
Institut: IBK
Departement: DBAUG

Project: BauhallenUhr.PrjPcb
Page: 3 from: 6

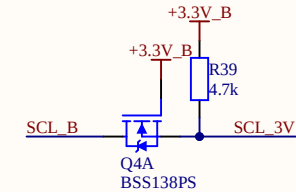
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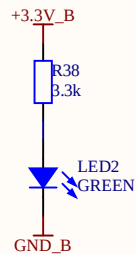
I2C Connector



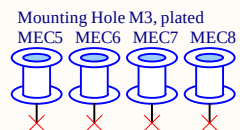
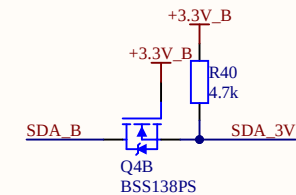
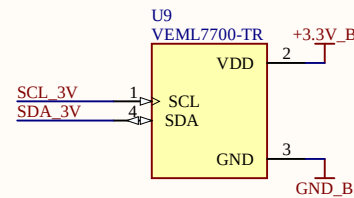
Shift with Pull - Ups



ON - LED



Brightness - Sensor



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Title of drawing: *Brightness_Sensor_Schematic.SchDoc*

Date: 30.07.2026 Time: 10:04:49

Revision: V0.1

Drawn by: Manuel Hueppi

Institut: IBK
Departement: DBAUG

Project: *BauhallenUhr.PrjPcb*

Page: 6 from: 6

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Board Stack Report